

Hákon Freyr Gunnarsson

AI engineer · Founder of Krates ehf.

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PROFILE

I build production AI systems that can be trusted with real work. With a BSc in mathematics, an MSc in AI and data science and two years as a statistician at deCODE genetics, I have shipped production systems in genomics, finance, travel and enterprise software, each time learning a new domain within months. Since March 2026 I have been building Krates: an on-premises AI department for Icelandic enterprises and the agent-first back-office kernels it runs on, developed by an engineering team of AI agents that I designed and lead. I trust measurements over demos.

EXPERIENCE

Founder · Krates ehf., Reykjavík

Mar 2026 – present

An AI-native back office for Icelandic enterprises, correct by construction: agents do the work, rules are executable logic, and a human consents before anything leaves the box. Now in pilots.

Sókrates – on-premises AI department

- One NixOS appliance (14 services in five credential-isolated containers) that ingests a company's systems, keeps a governed knowledge graph and lets named agents do back-office work over Slack, Teams, WhatsApp and email.
- A universal schema connector that compiles OpenAPI, JSON Schema, GraphQL and SQL DDL into typed Pydantic models and serves them to agents as MCP tools.
- Business rules as executable logic: Logica laws compile to DuckDB SQL, and each one returns the set of violations rather than a pass/fail, built from a library of hundreds of ratified rule primitives.
- Crash-safe agents: an agent that dies mid-task never repeats an external action, such as a message or a write to a customer's system (DBOS, write-ahead attempt records). Every governed action carries a signed consent token and is recorded as its own trace span in a queryable audit store.
- Agent evaluation against invariants over live systems (typed output, a selection from the valid set, the shape of the span tree) rather than string matching. Models are injected by configuration and compared on real workloads, from gateway-routed inference on a mini-PC to fully local inference on a DGX Spark.

Python · pydantic-ai · FastAPI · DBOS · Postgres · DuckDB/DuckLake · Logica · Logfire/OpenTelemetry · MCP · OAuth 2 · NixOS · SvelteKit

Krepis – back-office software designed for AI agents

- Business software assumes a human navigating screens. Krepis is built for agents working through APIs and logs, with the guarantees needed to handle money, payroll and other business operations.
- Seven kernels (double-entry accounting, payroll, workforce management, money movement, inventory, commitments and a work ledger), each limited to the critical core of its domain. They are event-sourced, append-only, idempotent and replayable, every action has a dry run, and balances are computed from the log rather than stored. An agent can preview any effect, retry safely, and trace every number back to the events behind it.
- Customer-specific business rules live above the kernels as logical predicates (Logica), so policy changes keep the same guarantees without touching kernel code. The kernels share one core and one Postgres, and each must pass the same conformance suite.
- Run by AI agents with no human supervisor: a fictional 12-employee company's September (180 shifts, payroll, inventory, 14 payments, a 27-transaction ledger) went through six kernels in one process. An independent checker found zero differences across 4,913 comparisons, payroll was exact to the króna, and a restart changed nothing.

The Weave – an engineering organisation staffed by AI agents

- I built the organisation that develops Krates: five named AI engineers (Claude and Codex), each with its own machine and role. They plan work, write code, review each other's changes, fix what the reviewers find and merge. Reviews always come from an agent other than the author.
- It runs day and night. The team files its own tickets, and a scheduler I built gives each task to the agent best placed to take it, weighing remaining subscription capacity, skills, availability and how much work it can run in parallel. AI subscriptions go in and reviewed production software comes out, at an average of 250 ISK (\$2) per merged change.
- Mistakes become procedure. A daily harvest turns what reviewers catch into 40 standard operating procedures holding 414 recorded lessons, which every agent loads before it touches the code they govern. The team's expertise compounds instead of resetting with each session.
- From July to mid-September 2026, the team averaged 106 merged changes a week. By September, 94% of code changes received independent review before merge, and reviewers found issues in 83% of them; only 0.14% of merged changes have ever been reverted. I act as CTO: architecture, engineering standards, and the decisions that need human judgement.

Backend Engineer · Wise lausnir (Microsoft Dynamics partner)

Oct 2024 – Mar 2026

- Designed and deployed Grimoire, an internal knowledge platform on Azure (Container Apps, Bicep IaC, Neo4j AuraDB) that Claude uses as an MCP server, rolled out to every technical employee through Claude Team admin. A full OAuth 2.0 stack (RFC 7591, 8414, 9728) lets colleagues sign in with their company email and nothing to install.
- Cut municipal onboarding from legacy on-premises systems into Fabric/OneLake from about three weeks to half a workday. Built the self-healing ETL framework for the Business Central solution using PySpark, a medallion architecture and models generated from OpenAPI schemas; its schema-driven design became the starting point for Sókrates's universal schema connector.

Data Scientist & Business Intelligence Lead · Travelshift

Aug 2022 – Sep 2024

Rebuilt the data infrastructure (a Kimball star schema in Snowflake, ETL across internal and external systems, Power BI) and the financial automation of Iceland's largest travel marketplace.

- Cash-flow forecasting model with a 0.42% margin of error on 2.4B ISK of volume over a six-month horizon.
- Automated payment reconciliation (Adyen, Teya and internal systems): unreconcilable transactions fell from 1 in 35 to under 1 in 1,000.
- Reduced the booking department from seven contractors to two while improving its performance. Started with no finance background and within months was producing annual reports that passed audit.

Data Analyst & Product Owner · Alfreð Atvinnuleit

May 2021 – Aug 2022

Product owner for the Giggó gig-economy platform (system architecture, data model). Built the BI environment with real-time dashboards. Wrote the grant proposal that secured 30M ISK from Rannís.

Instructor, Department of Computer Science · Reykjavík University

2018 – 2021

Programming, Data Structures, Calculus & Statistics, and Discrete Mathematics II, taught alongside the MSc.

Statistician & Bioinformatician · deCODE genetics (Amgen)

2015 – 2017

Designed a haplotype compression algorithm with more than 10× the compression of gzip, which let whole chromosomes be loaded into memory instead of in segments. Large-scale genetic data analysis.

SELECTED PROJECTS

Homegrown Hero Films · homegrownherofilms.com

AI video · GPU pipelines · YouTube & TikTok · 2026

Short films where a child is the hero. Behind them sits a video pipeline on rented GPUs, with my own [open-source extensions](#) to the MiniMax H3 video model, AI agents that review every film before release, upscaling and frame-interpolation finishing, and scheduled publishing.

Residual Modeling

Python · LightGBM · Postgres/PostGIS · FastAPI · Next.js · 2026

Icelandic home valuation that splits each sale price into multiplicative layers (structure, location, local market, liquidity and timing), fitted as residuals in log-price space, so every estimate shows where its value comes from. On 12,709 time-ordered out-of-fold sales the median error is 6.6%, and 5.9% for apartments.

Autopod

Go · Python/FastAPI · SQLite · Cloudflare R2 · speech-to-text · LLMs

Drop in a video and it is published to YouTube, Spotify and Apple Podcasts with subtitles in English, Icelandic and Polish, a summary, tags and hashtags, and clips of the most quotable moments. A Go control plane drives a stateless Python data plane through a crash-resilient job state machine; when encoding fails, an LLM proposes an ffmpeg fix that is validated before it runs. My first Go project, shipped to multi-tenant production.

SKILLS

AI & agents	pydantic-ai, MCP, Anthropic/OpenAI/Gemini APIs, agent evaluation, RAG and embeddings, durable agent workflows, tracing with Logfire/OpenTelemetry, local inference, diffusion models
Languages	Python (expert), SQL, TypeScript, Go, R, Logica
Data	Postgres, DuckDB/DuckLake, Snowflake, Neo4j, Fabric/PySpark, Kimball modelling, event sourcing, Power BI
Infrastructure	NixOS, Docker, Azure (Container Apps, Bicep), Cloudflare, Tailscale, GitHub Actions
Engineering	Domain-driven design, architecture enforced in CI (import-linter, tach), typed errors, uv/ruff/ty
Spoken	Icelandic (native), English (fluent)

EDUCATION

MSc Computer Science · Reykjavík University — AI and data science; thesis on recommendation systems

2017 – 2021

BSc Mathematics · University of Iceland — computational mathematics and computer science

2011 – 2016